Title: Detecting Online Transaction Fraud with Classification Algorithms

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Abstract

This study presents the development and evaluation of a machine-learning based approach to address the problem of detecting fraudulent online transactions. The objective of this research is to design a predictive classification model capable of improving accuracy, efficiency, and scalability compared to traditional methods.

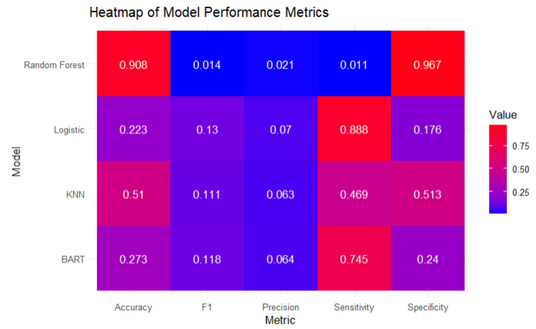
Introduction

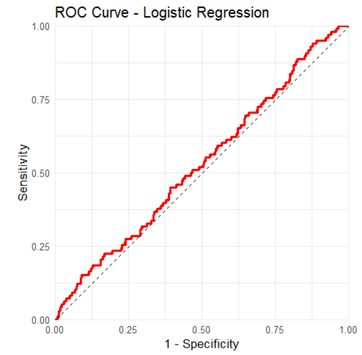
- The dataset is 7500 synthetic digital payment transactions labeled as either fraudulent or legitimate.
- Each row represents a single transaction and contains features about behavior, finance, device, and risk-related features..
- The goal is to find which model can most accurately detect a fraudulent transaction.

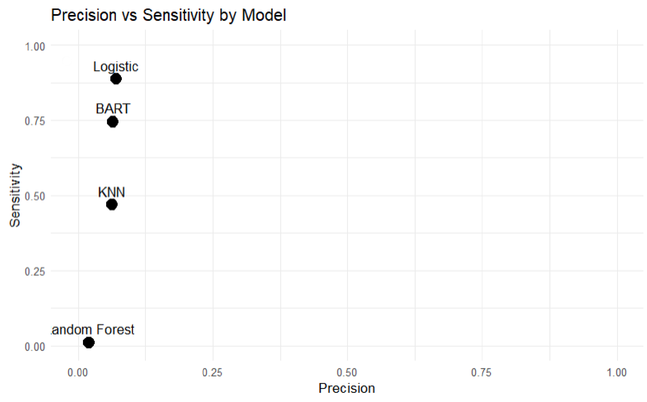
Methodology

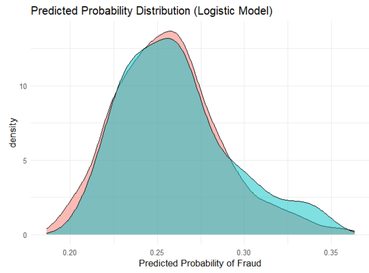
- Identify key variables and patterns through exploratory data analysis.
- Create classification models of various types (Logistic Regression, KNN, Random Forest, BART).
- Utilize validation methods, resampling, optimizing thresholds, data transformation techniques, stratification, and more to finely tune models.
- Select best model based on performance metrics (Accuracy, F-1, Specificity), provide clear logical reasoning why model is considered the best.

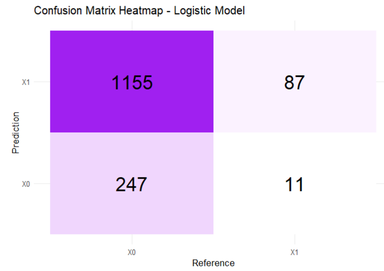
Results



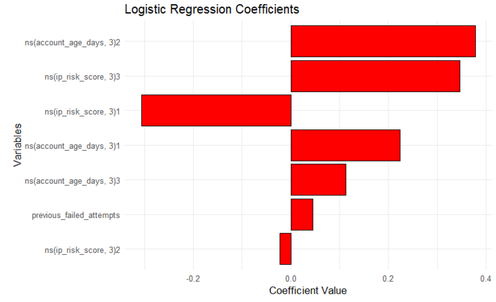












Model	Accuracy	Sensitivity	Specificity	Precision	F1
Logistic	**0.223**	**0.888**	**0.176**	**0.070**	**0.130**
KNN	0.510	0.469	0.513	0.063	0.111
Random Forest	0.908	0.011	0.967	0.021	0.014
BART	0.273	0.745	0.240	0.064	0.118

Conclusion

- Best model is Logistic with a sensitivity of 0.888 and F1 of 0.130
- Future studies may implement more sophisticated models (Neural Networks or Advanced Gradient Boosting)

References:

James, Gareth, et al. *An Introduction to Statistical Learning: with Applications in R*. 2nd ed., Springer, 2021.

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